# Tensor (Tensor) | Day 11 of ML |

stored data

↓

Why first tensor

→ tensor is nothing but data structure

→ Most Basic labery is - tensor.

Tensor is container → data stored in form of Number Ex- vector, metrix

→ Tensor and n-dimensional array are same

## What are tensor:-

2. 0D Tensor / scaler (zero D)

(2) (3)

scaler → 0D Tensor

Ex:-

```
import numpy as np
a = np.array(4)
```

output =

```
a
array(4)
a.ndim
= 0
```

3. 1D Tensor / vector.

[1, 2, 3, 4] → 1D Tensor

→ vector

→ 1D array

overall one D dimension है।

n.dim → 1

यह है तो one D dimension, But उसके जो vector है वह 4D dimension है → 4D

1D Tensor / vector

| Axis = dimension |

| No. of ~~rank~~ axis = rank = dimension |

Example

arr = np.array([1,2,3,4])

arr

→ arr([1,2,3,4])

arr.ndim

→ 1

[0,1,2,3]

↓

scaler → vector.

collection of vector

## 4. 2D Tensor / matrices:-

↓

matrix are nothing but collection of vectors.

[1,2,3] [4,5,6] [7,8,9]

$$\begin{bmatrix} [1,2,3] \\ [4,5,6] \\ [7,8,9] \end{bmatrix} \rightarrow 2D$$

Rank = 2 = ndim.

Ex:-

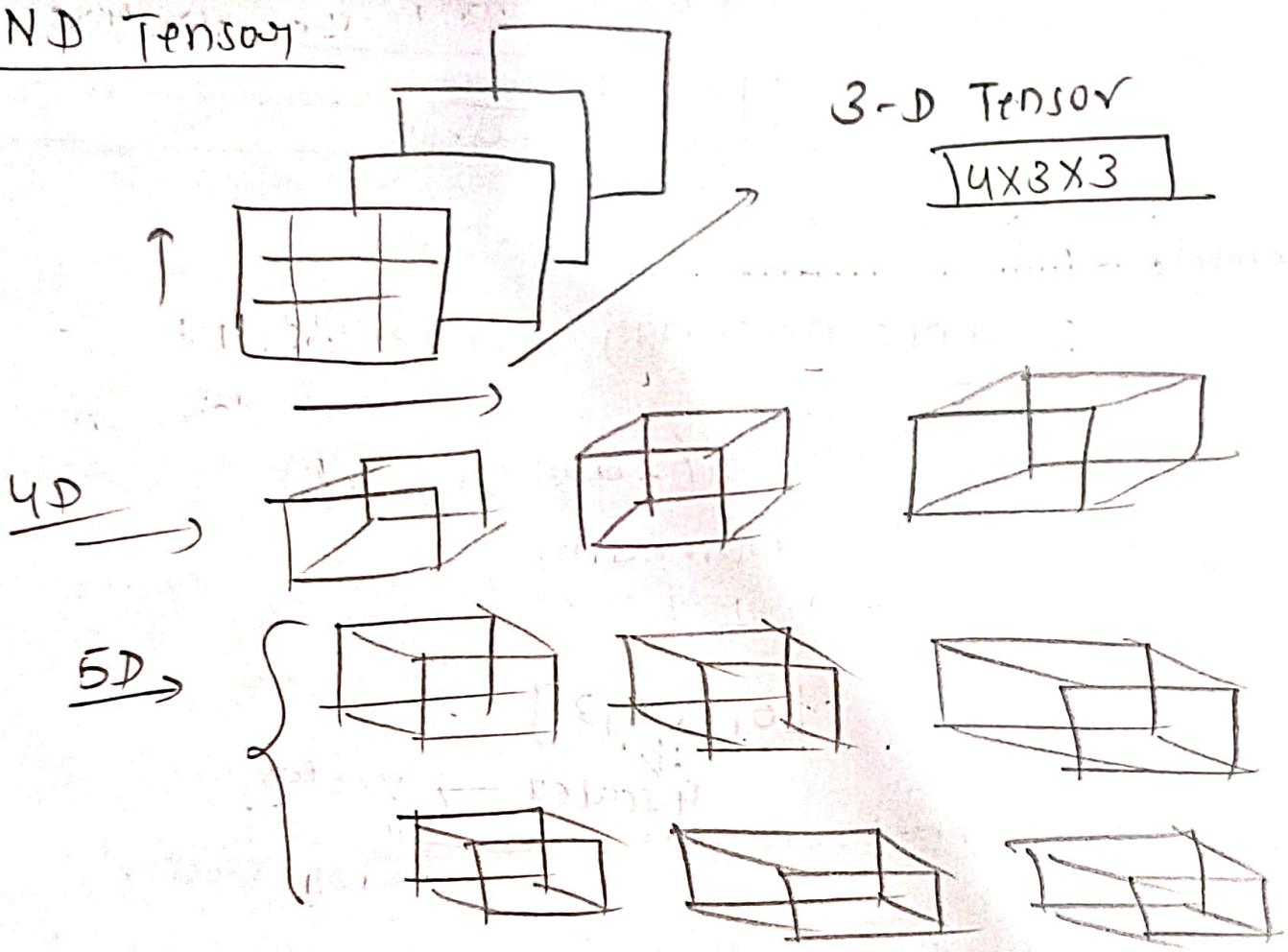
mat = np.array([[1,2,3],[4,5,6],[7,8,9]])

→ arr([ [1,2,3]
[4,5,6]
[7,8,9])

→ mat.ndim

→ 2

## 5. ND Tensor

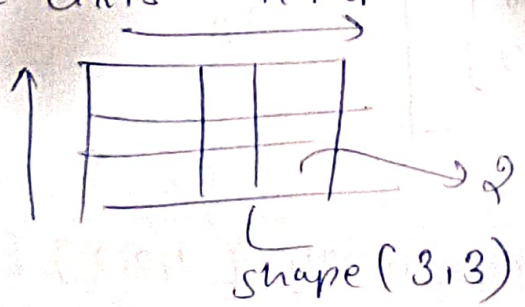


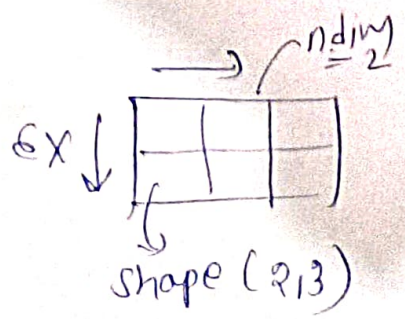

In machine learning may be only 0-5D

## 6. Rank, Axis, and shape.

No. of axis = Rank = No of dimension

shape



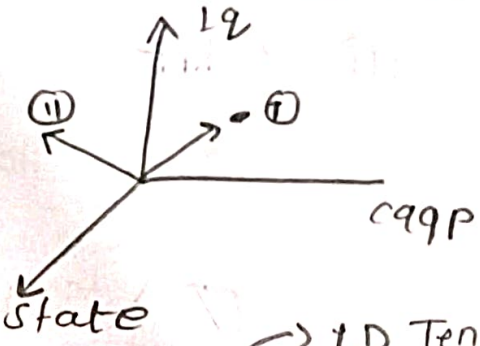
## 7. Example of 1D Tensor :- 1D Tensor / vector

Student dataset (10,000)

WB = 0
KR = 1

| cgpa | IQ | state | placement |
|---|---|---|---|
| 8.1 | 92 | WB = 0 | 1 |
| 7.2 | 102 | KR = 0 | 0 |
| | | | 1 |
| | | | 0 |
| | | | 1 |
| | | | 0 |
| | | | 1 |
| | | | 0 |
| | | | (10000) ∞ |

⇒ [8.1, 92, 0] ↳ 1D vector ↳ 3D vector




→ 1D Tensor

↳ [0, 1, 0, 1, 0, 1, 0] → 10000 num.

## 8. Example of 2D - Tensor :-

cgpa | iq | state | placement

[ [ — — — ] ] → vector
[ [ — — — ] ] → 2D

2-D tensor

[ ( — — — ) ]
[ ( — — — ) ]
[ ( — — — ) ]

2D tensor

placement

[ — ]
[ — ]
[ — ]

1D tensor.

## Example of 3D tensor :-

**NLP**

Hi Nitish
Hi Rahul
Hi Ankit

| Hi | Nitish | Rahul | Ankit |
|---|---|---|---|
| 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 |

$$= \Big[ \overset{1D}{[1,0,0,0]} \; \overset{1D}{[0,1,0,0]} \Big] = \text{2D Array}$$

$$\Rightarrow [1,0,0,0] \; [\,[0,0,1,0] = \text{2D Array}$$

$$\Rightarrow [1,0,0,0] \; [\,0,0,0,1] = \text{2D Array}\,]$$

— 3D array

So, there total Num of 2D array in there want that is — 3D Array

Bcz, collection of 2D array is called 3-D Array

Shape(3,2,4)

**Ex:-** Time-Series Data:-

Kisi Bhi stock ka highest or lowest price within day

— those doing daily until year.

Highest | lowest

Day-1 — — (365, 2) — 1st Toppers

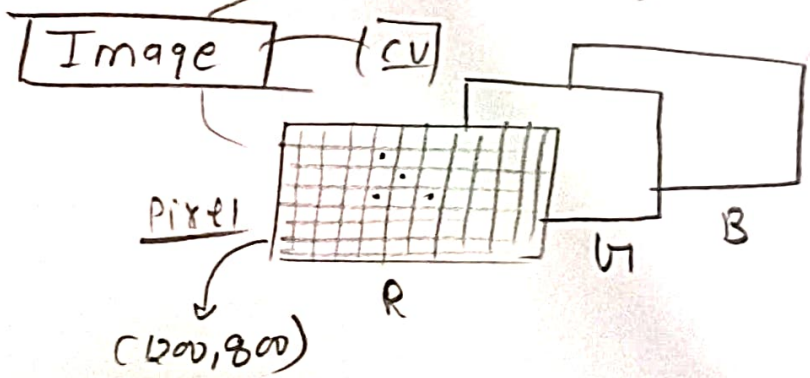
day-2

day-365 — — → 2D → 10 → (10, 365, 2) — time series

→ 3D Tensor

## Example of 4D Tensor

Image Based data:- Image collⁿ of fixed




Add like-3 (3, 1200, 800) → 3D Sensor

50 image (50, 1200, 800) → 4D Images

## Example of 5D Tensor:-

Viedo — collection of Image high speed transfer.

→ frame collection

4 viedo data → 60 sec → viedo → 30fps → 480 X 720 (3 channel RGB)

→ 5D tensor

1s-30 image → 1800 images

(480 X 720, 3)

5D → (4, (1800, 480, 720, 3))